Phase 3 (part 2): Layers and the Interaction Parameter

1.] Consider a network on nodes $\{A, B, C, D, E\}$ with two interaction layers:

$$A(\alpha) = A_{\text{strong}} + \alpha A_{\text{weak}}, \qquad \alpha \geq 0.$$

Below are the two weighted, undirected adjacency matrices.

$$A_{\text{strong}} = \begin{bmatrix} 0 & 4 & 2 & 0 & 1 \\ 4 & 0 & 3 & 2 & 0 \\ 2 & 3 & 0 & 1 & 2 \\ 0 & 2 & 1 & 0 & 5 \\ 1 & 0 & 2 & 5 & 0 \end{bmatrix} \qquad A_{\text{weak}} = \begin{bmatrix} 0 & 5 & 4 & 4 & 5 \\ 5 & 0 & 1 & 0 & 1 \\ 4 & 1 & 0 & 1 & 0 \\ 4 & 0 & 1 & 0 & 1 \\ 5 & 1 & 0 & 1 & 0 \end{bmatrix}$$

a.) Compute $A(0.5)$.

$$A(0.5) = \begin{bmatrix} 0 & 6.5 & 4 & 2 & 3.5 \\ 6.5 & 0 & 3.5 & 2 & 0.5 \\ 4 & 3.5 & 0 & 1.5 & 2 \\ 2 & 2 & 1.5 & 0 & 5.5 \\ 3.5 & 0.5 & 2 & 5.5 & 0 \end{bmatrix}$$

b.) For any α, compute the strength of nodes A and B, denoted by $s_A(\alpha)$ and $s_B(\alpha)$, respectively.

$$S_A(\alpha) = 7 + 18\alpha$$

$$S_B(\alpha) = 9 + 7\alpha$$

c.) For which values of α does node A have larger strength than node B?

$$S_A(\alpha) \geq S_B(\alpha)$$

$$7 + 18\alpha \geq 9 + 7\alpha$$

$$\Rightarrow \quad 11\alpha \geq 2$$

$$\boxed{\alpha \geq \frac{2}{11}}$$

$$\boxed{\alpha^* = \frac{2}{11}}$$

Critical value of interaction parameter.

2.] Let $V = \{1, 2, \ldots, 10\}$ and consider the complete graph on these nodes. We define two weighted layers:

- A_{strong}: for $i \neq j$,

$$w_{ij}^{(s)} = \begin{cases} 6 & \text{if } i, j \in S = \{1, 2, 3, 4\}, \\ 4 & \text{if } i, j \in T = \{5, 6, 7\}, \\ 2 & \text{if } i, j \in U = \{8, 9, 10\}, \\ 1 & \text{if } i \text{ and } j \text{ are in different groups.} \end{cases}$$

- A_{weak}: for $i \neq j$,

$$w_{ij}^{(w)} = \begin{cases} 5 & \text{if } i = 10 \text{ or } j = 10, \\ 1 & \text{otherwise.} \end{cases}$$

Define the combined network:

$$A(\alpha) = A_{\text{strong}} + \alpha A_{\text{weak}}, \qquad \alpha \geq 0.$$

a.) Compute the combined edge weight $w_{2,6}(\alpha)$ between nodes 2 and 6.

$$w_{2,6}(\alpha) = 1 + \alpha$$

b.) Compute the combined edge weight $w_{3,10}(\alpha)$ between nodes 3 and 10.

$$w_{3,10}(\alpha) = 1 + 5\alpha$$

c.) Compute the strength of node 1, node 8, and node 10 as functions of α.

$$S_1(\alpha) = w_{1,1}(\alpha) + w_{1,2}(\alpha) + \cdots + w_{1,10}(\alpha)$$
$$= 0 + 3(6+\alpha) + 5(1+\alpha) + (1+5\alpha) = 24 + 13\alpha$$
$$S_8(\alpha) = 7(1+\alpha) + (2+\alpha) + (2+5\alpha) = 11 + 13\alpha$$
$$S_{10}(\alpha) = 7(1+5\alpha) + 2(2+5\alpha) = 11 + 45\alpha$$

d.) Solve for the value(s) of α where $s_{10}(\alpha) = s_1(\alpha)$.

$$11 + 45\alpha = 24 + 13\alpha$$
$$32\alpha = 13$$
$$\boxed{\alpha = \frac{13}{32}}$$